

Software engineer with hands-on experience architecting and shipping production systems across full-stack web, cloud infrastructure, and AI/ML domains. Led end-to-end development of an aviation intelligence platform processing thousands of unstructured documents, and contributed to AI-powered call center and digital commerce products. Proven ability to design scalable systems using AWS, Docker, and serverless architectures while leading small engineering teams.

TECHNICAL SKILLS

Languages	JavaScript, TypeScript, Python, Java, PHP, SQL, HTML5, CSS3
Frontend	React, Next.js, React Native, Tailwind CSS, Ant Design, Material UI
Backend	Node.js, Express.js, Spring Boot, FastAPI, Django, Flask
AI / ML	LangGraph, LangChain, Gemini API, OpenAI API, PyTorch, Transformers
Databases	PostgreSQL, MySQL, MongoDB, Supabase, Redis, Sequelize ORM
Cloud & DevOps	AWS (Lambda, S3, ECS, EC2, Step Functions, IAM, CloudWatch), Docker, Kubernetes, Terraform, GitHub Actions, CI/CD
Tools & Practices	Git, NX Monorepo, ESLint, Husky, Conventional Commits, Agile/Scrum

PROFESSIONAL EXPERIENCE

Lead Engineer – Product Development | Sparengine

Present

Ascentis Technologies

sparengine.com

- Architected and engineered the entire aviation document intelligence platform from scratch, transforming thousands of pages of unstructured aircraft asset records into structured, queryable data
- Designed multi-level embedding pipelines to extract asset identity, maintenance events, and regulatory compliance indicators from large aviation dossiers
- Building multi-agent systems with LangGraph where specialized agents collaborate to deeply analyze and research aviation asset documents
- Established CI/CD pipelines, deployment automation, and infrastructure security across AWS (Lambda, Step Functions, ECS, S3, CloudWatch)
- Leading and mentoring a team of 3 engineers with code reviews, architecture guidance, and technical decision-making

Stack: Next.js, Node.js, Express, Supabase, LangGraph, LangChain, Gemini, AWS, Docker, Cloudflare Tunneling

Full-Stack Developer | Photos Ventoux

Ascentis Technologies

photos-ventouxsummit.fr

- Led development of a digital photo commerce platform where users select photos, convert them into products, and purchase digital/print versions with dynamic pricing
- Implemented multi-step gallery filter and rendering engine, accelerating page loads by 40% and improving search precision
- Built high-throughput media pipeline handling tens of gigabytes of ZIP archives — extraction, thumbnail generation via OpenAI API, Supabase S3 upload, and HD storage on AWS S3 — processing each image bundle in under one second
- Designed guest checkout flow with Stripe integration and quantity-based dynamic pricing logic
- Migrated platform from legacy PrestaShop to a unified Next.js codebase, simplifying deployments

Stack: Next.js, Tailwind, Supabase, AWS S3, Stripe, OpenAI API

Software Engineer Intern | DGloss – AI Call Center Platform

Softsora

- Developed user interfaces for operators and administrators including real-time management of operator availability status
- Built core backend APIs and business logic for call counting, per-agent/per-campaign call limit enforcement, and validation
- Implemented responsive, paginated data tables with server-side pagination and filtering on both frontend and backend
- Applied clean layered architecture (Models → DAOs → Services → Controllers → Routes) across the NX monorepo
- Worked with AWS Connect for contact center orchestration and Asterisk PBX for telephony integration

Stack: Next.js, Ant Design, Zustand, Express.js, Sequelize, PostgreSQL, AWS Connect, Asterisk PBX, Docker, NX

Software Engineer Intern | Insurance Event Management Platform

Softsora

- Built a web platform for a Japanese insurance company enabling event registration, online fee payment, and personalized QR code access cards
- Integrated Stripe payment gateway for secure online transactions and implemented QR code generation/scanning for event check-in
- Applied Japanese design principles emphasizing minimalism, clarity, and intuitive user flows

Stack: Full-Stack (Frontend & Backend), Stripe, Docker, QR Code Generation

PROJECTS

Email Signature Generator | Ascentis Technologies

- Designed and launched a signature builder with modular React components, slashing user setup time by 80% and standardizing branding across email clients
- Engineered cloud-powered storage and live-sync so signature updates propagate instantly across all devices

Stack: Next.js, React, TypeScript, Tailwind CSS, Supabase

RouteLead — Logistics Bidding Platform

- Built a mobile-first marketplace connecting empty-return drivers with parcel shippers, featuring bidding and offer selection
- Developed cross-platform mobile app with React Native and Spring Boot backend with Supabase auth/data layer

Stack: React Native, Spring Boot, Supabase, Docker

Stock API — Multi-Source Financial Data Platform

- Built a financial API serving real-time stock data from MarketWatch, Yahoo Finance, and Colombo Stock Exchange
- Developed unofficial CSE web scraping solution using Selenium and BeautifulSoup; implemented provider and factory design patterns

Stack: FastAPI, Python, Docker, Selenium, BeautifulSoup

Valerie Bourcaud Website + Headless CMS | Ascentis Technologies

- Built a modern website using Next.js with Notion as a headless CMS, cached via Upstash Redis
- Published reusable npm package: [notion-upstash-cms](#)

Stack: Next.js, Notion API, Upstash Redis, Tailwind

notion-upstash-cms — Published npm Package

- Created and published a reusable Node.js/TypeScript package enabling Notion as a CMS with Upstash Redis caching for fast content delivery in Next.js projects

Stack: Node.js, TypeScript, Notion API, Upstash Redis

RESEARCH

AVCCT-HCD: Harmful Content Detection in Short Videos

- Multi-label classification of harmful content (Adult, Suicide, Violent, Neutral) with a safety-first constraint of Recall $\geq 90\%$ for Suicide & Violent classes
- Designed two-stage cascaded bi-directional cross-attention architecture for progressive audio-video fusion using frozen VideoMAE + AST feature extractors
- Implemented 5-fold cross-validation with stratification, Wilcoxon + Bonferroni statistical tests, and comprehensive ablation studies
- Built production-ready, modular, and reproducible pipeline with Docker, ONNX, and TorchScript export support

Stack: PyTorch, PyTorch Lightning, Transformers, OpenCV, Librosa, FFmpeg, Docker, ONNX

EDUCATION

Bachelor of Science (Undergraduate), *Uva Wellassa University, Sri Lanka*

Present

Cumulative GPA: 3.72